

St02: assume that $p(m)$ is true and prove $p(m+1)$ we have:

$$0 < U_m < 1$$

$$0 < \frac{1}{2} U_m < \frac{1}{2} \quad (1)$$

$$0 < U_m < 1$$

$$1 < U_{m+1} < 2 \quad (2)$$

$1 \times 2 \Rightarrow$

$$0 < \frac{1}{2} U_m (U_{m+1}) < 1$$

$$0 < U_{m+1} < 1$$

So $p(m+1)$ is true by induction

we get $0 < U_m < 1$

2. Study the monotonicity of U_m :

$$U_{m+1} - U_m = \frac{1}{2} U_m (U_m + 1) - U_m$$

$$= \frac{1}{2} U_m^2 + \frac{1}{2} U_m - U_m$$

$$= \frac{1}{2} U_m^2 - \frac{1}{2} U_m < 0$$

$$U_{m+1} - U_m = \frac{1}{2} U_m (U_m - 1)$$

$$0 < U_m < 1 \Rightarrow \begin{cases} U_m > 0 \\ U_m - 1 < 0 \end{cases}$$

$$\text{So } U_{m+1} - U_m < 0$$

then $U_m \searrow$

3. deduce that U_m is c.v

we have U_m is decreasing and bounded below so it is c.v and its limit is:

$$\lim U_m = l$$

$$U_{m+1} = f(U_m) \Rightarrow \frac{1}{2} x(x+1)$$

$$f(l) = l$$

$$\Rightarrow \frac{1}{2} l(l+1) = l$$

$$\frac{1}{2} l^2 + \frac{1}{2} l - l = 0$$

$$\frac{1}{2} l^2 - \frac{1}{2} l = 0 \quad \begin{cases} l_1 = 0 \\ l_2 = 1 \end{cases}$$

$$\text{So } \lim U_m = 0$$

II Suppose $a > 1$

1) prove that $U_m > 1 \Rightarrow p(m)$

by induction:

St0: for $m_0 = 0 \Rightarrow U_0 > 1 \Rightarrow p(0) \vee$

St1: assume that $p(m)$ true and prove $p(m+1)$:

$$U_m > 1$$

$$\frac{1}{2} U_m > \frac{1}{2} \quad (1)$$

$$U_m > 1$$

$$U_{m+1} > 2 \quad (2)$$

$$\frac{1}{2} U_m (U_m + 1) > 1$$

$p(m+1)$ is true $\Rightarrow p(m)$ true

1) Show that U_n is $\nearrow$

$$\begin{aligned} U_{n+1} - U_n &= \frac{1}{2} U_n (U_n + 1) - U_n \\ &= \frac{1}{2} U_n (U_n - 1) \end{aligned}$$

$$\begin{cases} U_n > 1 \\ \frac{1}{2} U_n > \frac{1}{2} > 0 \end{cases} \Rightarrow \begin{cases} U_n > 1 \\ U_{n-1} > 0 \end{cases}$$

So $U_{n+1} - U_n > 0 \Rightarrow U_n \nearrow$

o prove that $U_n > \left(\frac{1+a}{2}\right)^n$

by induction:

for $n=0 \Rightarrow U_0 = a > 1 = \left(\frac{1+a}{2}\right)^0$

assume $U_n > \left(\frac{1+a}{2}\right)^n$

Prove $U_{n+1} > \left(\frac{1+a}{2}\right)^{n+1}$

$$\frac{1}{2} U_n (U_n + 1) > \left(\frac{1+a}{2}\right)^{n+1}$$

$U_0 = a > 1$ and $U_n \nearrow \Rightarrow U_n > U_0$

2) deduce the nature of $(U_n)_n$

$$\ln U_{n+1} > \ln \left(\frac{1+a}{2}\right)^{n+1} = +\infty$$

So U_n is d.V

EX03:

$$f(x) = \begin{cases} e^x \left(-x + \frac{4}{\pi}\right) & x \leq 0 \\ \frac{\sin(x) - \cos(x)}{x - \frac{\pi}{4}} & 0 < x < \frac{\pi}{4} \\ \sqrt{2} + \frac{\pi}{4} - x & x \geq \frac{\pi}{4} \end{cases}$$

1) determine the Set D_f :

$$D_{f_1} =]-\infty, 0]$$

$$D_{f_2} =]0, \frac{\pi}{4}[$$

$$D_{f_3} =]\frac{\pi}{4}, +\infty[$$

$$\left. \begin{matrix} D_{f_1} \\ D_{f_2} \\ D_{f_3} \end{matrix} \right\} D_f = D_{f_1} \cup D_{f_2} \cup D_{f_3}$$

$$\Rightarrow D_f =]-\infty, \frac{\pi}{4}[\cup]\frac{\pi}{4}, +\infty[$$

$$D_f = \mathbb{R} \setminus \left\{ \frac{\pi}{4} \right\}$$

2) Study the continuity on D_f :

o on $]-\infty, 0]$ $f(x)$ is continuous
cuz it's a combination of a continuous function $(e^x / -x + \frac{4}{\pi})$

o on $]0, \frac{\pi}{4}[$ $f(x)$ is continuous
cuz it's a combination of CN functions

o on $]\frac{\pi}{4}, +\infty[$ $f(x)$ is continuous
cuz it's a polynomial

0 at $x_0 = 0$

$$f(0) = \frac{4}{\pi}$$

$$\begin{aligned}\lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} \frac{\sin(x) - \cos(x)}{x - \frac{\pi}{4}} \\ &= \lim_{x \rightarrow 0^+} \frac{-1}{0 - \frac{\pi}{4}} = \frac{4}{\pi}\end{aligned}$$

$$\begin{aligned}\lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} e^x \left(-x + \frac{4}{\pi}\right) \\ &= \frac{4}{\pi}\end{aligned}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

then f is continuous on D_f

Study the Diff:

$$\begin{aligned}f'_+(0) &= \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{\sin(x) - \cos(x)}{x - \frac{\pi}{4}} - \frac{4}{\pi}}{x}\end{aligned}$$

with l'Hopital's we find

$$\left(1 - \frac{4}{\pi}\right) \left(\frac{16}{\pi^2}\right)$$

$$\begin{aligned}f'_-(0) &= \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0^-} e^x \left(-x + \frac{4}{\pi}\right) - \frac{4}{\pi} = -1\end{aligned}$$

$f'_+(0) \neq f'_-(0)$ so f is not diff at $x_0 = 0$

3) does f accept a continuous extension?

we have $D_f = \mathbb{R} \setminus \{\frac{\pi}{4}\}$

f is not defined at $\frac{\pi}{4}$ so f extends to a continuous function

$$\Leftrightarrow \lim_{x \rightarrow \frac{\pi}{4}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{4}^-} f(x) = ?$$

$$\lim_{x \rightarrow \frac{\pi}{4}} f(x) = \lim_{x \rightarrow \frac{\pi}{4}} \sqrt{2} + \frac{\pi}{4} - x = \sqrt{2}$$

$$\lim_{x \rightarrow \frac{\pi}{4}} f(x) = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{x - \frac{\pi}{4}} = \frac{0}{0}$$

$$\text{Let } y = x - \frac{\pi}{4} \text{ if } x \rightarrow \frac{\pi}{4} \Rightarrow y \rightarrow 0$$

$$\begin{aligned}\lim_{y \rightarrow 0} \frac{\sin(y + \frac{\pi}{4}) - \cos(y + \frac{\pi}{4})}{y} &= \sqrt{2} \lim_{y \rightarrow 0} \frac{\sin y}{y} \\ &= \sqrt{2}\end{aligned}$$

So f admits a continuous extension given by

$$g(x) = \begin{cases} f(x) & ; x \in \mathbb{R} \setminus \{\frac{\pi}{4}\} \\ \sqrt{2} & ; x = \frac{\pi}{4} \end{cases}$$



Series n° 2 - Numerical Sequences-

Generalities

Exercise 1. Comment if the following statements are true or false? And give a valid reason for saying so.

- Every bounded sequence is convergent.
- Every sequence $(u_n)_n$ satisfies $u_{n+1} \geq u_n$ is increasing.
- The sum of two divergent sequences is a divergent sequence.
- The product of two divergent sequences is a divergent sequence.
- If the sequence $(u_n)_{n \in \mathbb{N}}$ is increasing and $u_n \leq n$, then $(u_n)_n$ is convergent.
- If $\lim_{n \rightarrow +\infty} u_n^2 = \ell^2$, then $\lim_{n \rightarrow +\infty} u_n = \ell$.
- If $\lim_{n \rightarrow +\infty} |u_n| = |\ell|$, then $\lim_{n \rightarrow +\infty} u_n = \ell$.

$U_m = m - \frac{1}{m} + U_m \leq m$
 $\lim U_m = +\infty$ div



Convergent Sequences

Exercise 2. I) Consider the real sequence $(u_n)_{n \in \mathbb{N}^*}$ defined as: $u_n = \frac{n^2 + 2n}{4n^2 + 1}$.

$\lim_{n \rightarrow +\infty} (u_n)_n$ converges to $\ell \iff (\forall \varepsilon > 0, \exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N}; n \geq n_0 \Rightarrow |u_n - \ell| < \varepsilon)$.

Find the absolute value of this inequality $0.24 < u_n < 0.26$, then deduce ℓ, ε and n_0 .

► Use the previous definition to prove each of the following:

$$\lim_{n \rightarrow +\infty} \frac{2 \ln(n+1)}{\ln(n)} = 2;$$

$$\lim_{n \rightarrow +\infty} \frac{n^3 + 1}{n^3 + n^2 + 2} = 1.$$

II) Study the convergence of the following sequences:

1) $u_n = \frac{\cos(\sqrt{n})}{\sqrt[3]{n}}$,

2) $u_n = \frac{2n}{n^3 + 1} \sum_{k=0}^n k, n \in \mathbb{N}$,

3) $u_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}$,

4) $u_n = \frac{\alpha^n - \beta^n}{\alpha^n + \beta^n}$,

5) $u_n = \frac{1}{n^2} \sum_{k=1}^n [kx], n \in \mathbb{N}^*$,

6) $u_n = \sqrt[3]{n+1} - \sqrt[3]{n}$,

7) $u_n = \left(\frac{n^2 - n + 3}{n^2 + 3n - 1} \right)^n$, .. 8) $u_n = \frac{5 \times 7 \times 9 \cdots (2n+5)}{4 \times 7 \times 10 \cdots (3n+4)}$.

$\lim a^n \begin{cases} 0 & \text{if } |a| < 1 \\ +\infty & \text{if } a > 1 \end{cases}$

Adjacent Sequences

Exercise 3. I) Let the following recursive sequences:

$$\begin{cases} u_0 = 0 \\ u_{n+1} = \frac{u_n + 1}{u_n + 2}, \quad n \in \mathbb{N}, \end{cases}$$

$$\begin{cases} v_0 = 2 \\ v_{n+1} = \frac{v_n + 1}{v_n + 2}, \quad n \in \mathbb{N}. \end{cases}$$

(a) Show that: $u_{n+1} = 1 - \frac{1}{u_n + 2}$.

(b) Prove that: $\forall n \in \mathbb{N}; v_n \geq u_n \geq 0$.

(c) Study the monotonicity of $(u_n)_n$ and $(v_n)_n$.

(d) Confirm that: $\forall n \in \mathbb{N}; v_{n+1} - u_{n+1} \leq \frac{1}{4}(v_n - u_n)$.

(e) Deduce that the sequences $(u_n)_n$ and $(v_n)_n$ are adjacent.

(f) Determine $\sup(u_n)$, $\inf(u_n)$, $\sup(v_n)$, $\inf(v_n)$, $\max(v_n)$, and $\min(v_n)$.

II) Consider the sequence of general term: $v_n = 1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{(-1)^{n+1}}{n}$, $\forall n \in \mathbb{N}^*$.

► Prove that the two subsequences $(v_{2n})_n$ and $(v_{2n+1})_n$ are adjacent, then deduce the nature of $(v_n)_n$.

Divergent Sequences

Exercise 4. I) Consider the real sequence $(u_n)_{n \in \mathbb{N}^*}$ defined as: $u_n = n\sqrt{n}$.

$$(u_n)_n \text{ diverges to } \pm \infty \iff \left(\begin{array}{l} \forall A > 0, \exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N}; n \geq n_0 \Rightarrow u_n > A \\ \forall A > 0, \exists n_0 \in \mathbb{N}, \forall n \in \mathbb{N}; n \geq n_0 \Rightarrow u_n < -A \end{array} \right).$$

Explain the definition that corresponds to the limit of $(u_n)_n$, such that $u_n \in (10^6; +\infty)$, then deduce the values of A and n_0 .

► Use the previous definitions to prove each of the following:

$$\lim_{n \rightarrow +\infty} \ln(\ln(2^{n+1})) = +\infty;$$

$$\lim_{n \rightarrow +\infty} \frac{1}{2}(e^{\frac{1}{n}} - n) = -\infty.$$

II) Choose the suitable way (subsequences, comparison test) to show that the below sequences are divergent:

$$1) u_n = \cos\left(\frac{n\pi}{4}\right), \quad 2) u_n = \left(1 + \frac{1}{n}\right)^{n^2}, \quad 3) u_n = \frac{n + (-1)^n n}{n - (-1)^n \frac{n}{2}}, \quad 4) \frac{n! + 3^n}{n - 2^n}.$$

Cauchy Sequences

Exercise 5. I) Consider the real sequence $(u_n)_{n \in \mathbb{N}^*}$, such that:

$$|u_{n+1} - u_n| \leq k|u_n - u_{n-1}|, \quad n \in \mathbb{N}^*, \quad k \in (0; 1).$$

(a) Show that: $\forall n \in \mathbb{N}^*, |u_{n+1} - u_n| \leq k^{n-1}|u_2 - u_1|$;

(b) Prove that: $\forall p, q \in \mathbb{N}^*, q < p$, we have $|u_p - u_q| \leq \frac{k^{q-1}}{1-k}|u_2 - u_1|$;

(c) Deduce the nature of $(u_n)_n$.

II) In view of the Cauchy criterion, determine the nature of the sequences $(u_n)_n$ defined by:

$$u_n = \sum_{k=1}^n \frac{1}{\sqrt{k}}$$

$$u_n = \sum_{k=1}^n \frac{\sin k}{2^k}.$$



$$\forall \varepsilon > 0, \exists m_0 \in \mathbb{N}, \forall p, q \in \mathbb{N}$$

$$p > q > m_0 \Rightarrow |u_p - u_q| < \varepsilon$$

$$\sin()$$

$$\cos() \leq 1$$

$$(-1)^n$$

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Series n° 3 -Functions, Limits and Continuity-

Functions

Exercise 1. 1. Compute without using a calculator :

1) $\sin(\arcsin(1) - \arccos(\frac{\sqrt{3}}{2}))$, 2) $\arctan(-\sqrt{3})$, 3) $\arcsin(2 \cos(\frac{2\pi}{3}))$, 4) $2 \operatorname{arccot}(-1)$.

2. Simplify the following expressions:

1) $\cos(\arcsin(x))$, 2) $\operatorname{ch}(\operatorname{argsh}(x))$, 3) $\tan(\arcsin(x))$, 4) $\operatorname{ch}(\operatorname{argth}(x))$.

3. Prove that: 1) $\forall x \in [-1, +1]$, $\arcsin(x) + \arccos(x) = \frac{\pi}{2}$.

2) $\forall x \in (-1, 1)$, $\operatorname{argth}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$.

Exercise 2. Find the domain of each function using interval notation:

1) $f(x) = 2 \arctan \left(\sqrt{\frac{1}{x} - 1} \right) + \arcsin(2x - 1)$, 2) $f(x) = (1 + \ln(x))^{\frac{1}{x}}$, 3) $f(x) = \arccos \left(\frac{2x}{1+x^2} \right)$,

4) $f(x) = \sqrt{\frac{x^2 + 2x + 3}{x^2 - 1}}$, 5) $f(x) = \operatorname{argch} \left(\frac{1}{2} \left(x + \frac{1}{x} \right) \right)$, 6) $f(x) = \frac{1}{\sqrt{\sin x}}$, 7) $f(x) = \frac{x^3 + 3}{1 - |x|}$.

Exercise 3. 1) Examine the parity of each of the given function:

1) $f(x) = \ln(x + \sqrt{1+x^2})$, 2) $f(x) = \arctan \left(\frac{\arcsin(2x\sqrt{1-x^2})}{x^3 \operatorname{sh} x} \right)$, 3) $f(x) = x \frac{a^x - 1}{a^x + 1}$ ($a > 0$).

II) Show that the following functions are periodic and compute their smallest period:

1) $f(x) = \sin^2 x$, 2) $f(x) = \frac{\cos 5x}{\sin 3x}$.

III) Prove that f is 2π -periodic, what is the parity of f ? Deduce a study interval I .

$f(x) = \arcsin(1 - 2 \cos^4(x))$.

Limits

Exercise 4. Calculate the following limits:

1) $\lim_{x \rightarrow +\infty} \left(\frac{x-1}{x+1} \right)^x$, 2) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$, 3) $\lim_{x \rightarrow 0} \frac{x}{a} \left[\frac{b}{x} \right]$ ($a, b \in \mathbb{R}^* \times \mathbb{R}$),

4) $\lim_{x \rightarrow -\infty} \frac{x^4 + 1}{\cot \frac{1}{x}}$, 5) $\lim_{x \rightarrow +\infty} \frac{2 \operatorname{ch}^2(x) - \operatorname{sh}(2x)}{x - \ln(\operatorname{ch}(x)) - \ln(2)}$, 6) $\lim_{x \rightarrow a} \frac{\sqrt{x} - \sqrt{a} + \sqrt{x-a}}{\sqrt{x^2 - a^2}}$, = $\frac{1}{\sqrt{2a}}$

7) $\lim_{x \rightarrow 0} \frac{\ln \sqrt{x}}{\sqrt[3]{x}}$, 8) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin(2x) \sin(x - \frac{\pi}{4})}{\sin x - \cos x}$, 9) $\lim_{x \rightarrow 2} \frac{\arctan(x-2)}{x^2 - 4}$.

Exercise 5. I) Prove the following limits using their precise definition:

1) $\lim_{x \rightarrow -2} \frac{x^2 + x + 1}{1 - x} = 1$, 2) $\lim_{x \rightarrow 1} \frac{x + 2}{x - 1} = -\infty$, 3) $\lim_{x \rightarrow -\infty} \frac{1}{x^3} = 0$.

II) Choose two ways to prove that the following functions have no limits at x_0 .

1) $f(x) = \sin(x)$, $x_0 = +\infty$, 2) $f(x) = x \sqrt{1 + \frac{1}{x^2}}$, $x_0 = 0$.

Continuity

Exercise 6. I) Study the continuity of the following functions:

$$1) f(x) = (x + [\cos(x)])^2, \forall x \in [0, \pi], \quad 2) f(x) = \begin{cases} x^p \sin \frac{1}{x} & \text{if } x \neq 0 \quad (p \in \mathbb{N}); \\ \alpha & \text{if } x = 0 \quad (\alpha \in \mathbb{R}). \end{cases}$$

II) Find the values of a and b that makes each function continuous over the given interval.

$$f(x) = \begin{cases} (x-1)^2 & \text{if } x \leq 0; \\ ax+b & \text{if } 0 < x < 1; \\ \sqrt{x} & \text{if } x \geq 1. \end{cases} \quad f(x) = \begin{cases} -2 \sin x & \text{if } x \leq -\frac{\pi}{2}; \\ a \sin x + b & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2}; \\ \cos x & \text{if } x \geq \frac{\pi}{2}. \end{cases}$$

Exercise 7. Determine if the given functions can be extended to the continuous functions:

$$1) f(x) = \frac{1}{x} \sin\left(\frac{1}{x}\right), \quad 2) f(x) = \sin(x+1) \ln|x+1|, \\ 3) f(x) = \frac{1}{1-x} - \frac{2}{1-x^2}, \quad 4) f(x) = \begin{cases} \frac{\sqrt{x} - \sqrt{2} + \sqrt{x-2}}{6\sqrt{x^2-4}} & \text{if } x > 2; \\ \frac{x^3 - 3x + 2}{x^3 - 8} & \text{if } x < 2. \end{cases}$$

Intermediate Value Theorem (IVT)

Exercise 8. I) Consider the function f defined by:

$$f(x) = x \ln(x) - 1, \forall x \in [1, 2].$$

Prove that f has a root between 1 and 2.

II) Let $a < c < d < b$ and f be a function defined on $[a, b]$. Show that:

If f is continuous on $[a, b]$ and $f(c) + f(d) = k$, then there exists $y_0 \in [a, b]$ such that $f(y_0) = \frac{k}{2}$.





Series n° 4 -Differentiation on $\mathbb{R}$ -

Exercise 1. • Find the derivative of each function:

$$1) f(x) = \ln |x - e^x \tan(x)|, \quad 2) f(x) = \sin(\cos(\sin(x))), \quad 3) f(x) = \arccos\left(\frac{2\sqrt{x}}{1+x}\right).$$

• Show that:

$$1) \forall x \in (-1, 1) : (\arcsin(x))' = \frac{1}{\sqrt{1-x^2}}, \quad 2) \forall x \in (-1, 1) : (\operatorname{argth}(x))' = \frac{1}{1-x^2}.$$

Exercise 2. Using L'Hospital's rule, calculate the following limits:

$$1) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin(x))}{(\pi - 2x)^2}, \quad 2) \lim_{x \rightarrow 1} \frac{\arctan\left(\frac{x^2-1}{x^2+1}\right)}{x-1}, \quad 3) \lim_{x \rightarrow +\infty} x \left(\left(1 + \frac{1}{x}\right)^x - e \right).$$

Exercise 3. Study the differentiability of the following functions:

$$1) f(x) = \begin{cases} \frac{\sin^2(\pi x)}{x - \pi}, & x \neq \pi \\ 0, & x = \pi \end{cases} \quad 2) g(x) = \begin{cases} \arctan(x), & |x| \leq 1, \\ \frac{\pi|x|}{4x} + \frac{x-1}{2}, & |x| > 1. \end{cases}$$

Exercise 4.

1. Give an example of a continuous function from $\mathbb{R}$ to $\mathbb{R}$ which is differentiable only on $\mathbb{R}^*$.
2. Consider the following function:

$$f_\alpha(x) = \begin{cases} |x|^\alpha \cos\left(\frac{1}{|x|}\right), & \alpha > 0, x \neq 0, \\ 0, & x = 0. \end{cases}$$

- Show that the function f_α is continuous on $\mathbb{R}$.
- Study the differentiability of f_α .
- Calculate the derivative function f'_α .

Exercise 5.

- Let function f be defined as $f(x) = \frac{a}{bx+c}$, $a, b, c \in \mathbb{R}^*$

1. Find the n^{th} derivative of f .

2. Deduce the 10^{th} derivative of $g(x) = \frac{2}{1-3x}$.

- Using Leibniz's formula, calculate $h^{(3)}$ for $h(x) = x^2 e^x$.

Exercise 6.

1. State the mean value theorem for $f(x) = \arctan(x)$ on $[a, b]$.
2. Use the above theorem to prove the following inequality:

$$\forall a, b \in \mathbb{R}, 0 \leq a < b: \frac{b-a}{1+b^2} < \arctan(b) - \arctan(a) < \frac{b-a}{1+a^2}.$$



Exercise 7.

1. State Rolle's theorem on $[a, b]$.
2. Can be applied Rolle's theorem in the following cases?

$$a) f(x) = \begin{cases} 3 - x^2, & x \leq 1, \\ \frac{1}{x}, & x > 1, \end{cases} \text{ on } [0, 2], \quad b) g(x) = \sqrt[3]{(x-2)^3} \text{ on } [0, 4].$$

Exercise 8.

1. State the Mac-Laurin-Lagrange formula at order n .
2. Let the following function:

$$f(x) = e^x$$

- Calculate the n^{th} derivative of f .
- Give the Mac-Laurin-Lagrange formula of f at order n .

3. Applications:

- Show that:

$$1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} < e < 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} + \frac{e}{(n+1)!}.$$

- Let the sequence $(u_n)_n$ of general term

$$u_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}.$$

Calculate $\lim_{n \rightarrow +\infty} u_n$.